04 Tracking, Time resolved microscopy.

Assignment 4, 25 points in total; 16.6% of the total score for the practical part.

In this assignment the problem of feature extraction from a time-series is addressed. Researchers would like to understand the moving speed, distance trajectory and shape changes of cells. In the experiment presented for this assignment, the cells are cancer cells. The images of the cells are given to you as is. Let us assume that the time-lapse procedure takes a snapshot of the cells at intervals of **2 minutes**. The living conditions of the cells are kept constant (temperature, humidity, oxygen-flow) so that they will survive well over a longer period in culture.

We will use the following series of images:

(A) MTLn3-Ctrl0000.tif – MTLn3-Ctrl0029.tif, and

(B) MTLn3+EGF0000.tif – MTLn3+EGF0029.tif

Series A is the control, whereas in the experiment in series B, a growth factor was added to the cells, and this might resort in an effect on the movement and shape of the cells. We can measure this effect through a readout in the images. Henceforth, Series A will be referred to as the Control Series, and Series B will be referred to as the Experimental Series.

This assignment has to be completed using PyDIP.

Part 1

1. In the given sequence (00-29) pick 15 cells in the Control Series. These cells are traced over the sequence of the 30 images. Indicate your choices with a label/number in the initial image. These labels are the result from a segmentation operation that successfully finds the relevant cells in the image(s). Develop, apply, explain and motivate your segmentation procedure. (3)
2. Develop an algorithm that can trace the cells over the time-lapse sequence. It is important to list a criterion or criteria by which you can determine that a cell in a next frame is the same cell in the current frame. Your algorithm is based on that criterion, or those criteria. This should be clearly stated in the explanation of the algorithm. (3)
3. From the 15 cells that are selected, do a manual check of the tracing algorithm (discussed in the previous step) on 5 cells in the Control Series and provide a visualization of the traces. Explain your validation procedure – basically, this is a ground truth procedure that provides insight for the success of an algorithm. (2)
4. Apply the algorithm to the chosen 15 cells in the Control Series, and 15 cells (picked in the same manner) in the Experimental Series. This will result in traces of segmented cells that are extracted over the sequence. Provide appropriate visualizations of the traces. (4)

Part 2

Shape and texture of the cells may vary over time and over the different conditions. For shape you can start by computing the area and perimeter. However, as cells may stretch, in addition, compute the roundness as shape feature. As from the segmentation, a mask for each cell in the tracking is obtained, now compute texture features for each cell using the cell mask; i.e., mean, standard deviation, smoothness and uniformity (refer to the additional document on Brightspace for more information on Uniformity).

1. From the cells obtained from the tracking, compute the aforementioned features for shape and texture. This is for both conditions and presented in two different tables. (4)
2. Compute for both conditions, the cell velocity and distance trajectory and present this in a table. Again, we assume a time interval of 2 minute between the images. (2)
3. Within the cells that you have followed, identify differences in the trajectories by assessing the trajectories separately for both series. (2)
4. From the size, shape and texture of the Control and Experimental Series and deduce possible differences between the two series. (2)
5. Study the correlation between speed with shape and texture and visualize it. Interpret the correlation strength you found. (2)
6. Comment on image quality and image resolution in all dimensions studied. (1)

In this tracking experiment you should document your results with the intermediate images. These images can then be referred to in the tables that you provide with the measurements. Document your algorithm well with comment lines. Please submit the code as a separate .ipynb or .py file.